

PcapPoisoning

 **Security Warning:** Work in an isolated environment (virtual machine or container). Never open unknown files on your primary machine.

Challenge Description

How about some hide and seek heh?

Download this file and find the flag.

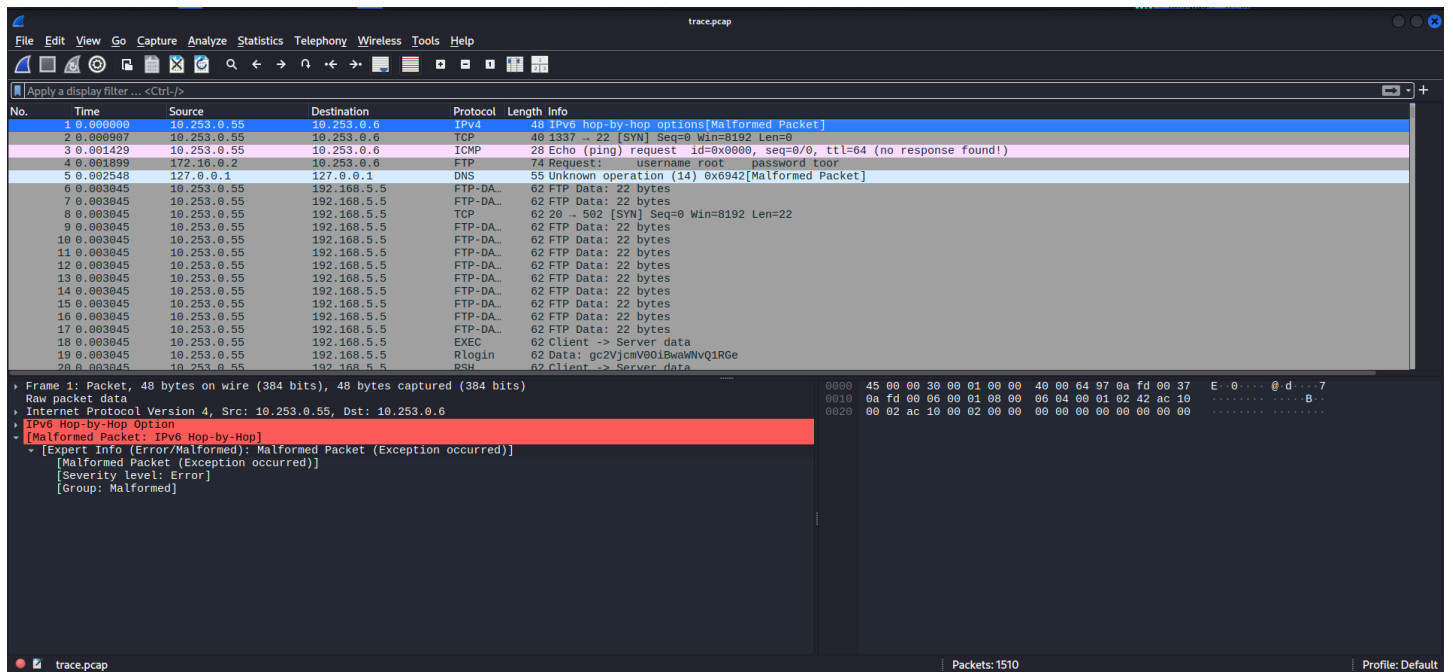
Tools Used

- `file` – Determine file type
 - `strings` – Extract printable character sequences from binary files
 - `grep` – Filter output for specific patterns
 - `tshark` – Command-line Wireshark tool for packet analysis
 - `sort` – Organize timestamps in chronological order
 - `cut` – Extract specific fields from output
 - `xxd` – Convert hexadecimal to binary data
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Complete Solution

Step 1: Download and Prepare

Download the PCAP file named `trace.pcap` from the challenge page.



Step 2: Basic File Inspection

First, verify the file type and general information:

```
file trace.pcap
```

Expected output:

```
trace.pcap: pcap capture file, microsecond ts (little-endian) - version 2.4 (Raw IPv4,
capture length 65535)
```

The file is a network packet capture (PCAP) containing raw IPv4 traffic.

Step 3: Quick String Search

Sometimes flags are directly visible as plain text strings within the binary data. Let's try a simple `strings` scan:

```
strings trace.pcap | grep -i "picoCTF"
```

Output:

picoCTF{<REDACTED>}9~

Key finding: The flag appears directly in the output! The `strings` command extracts readable ASCII sequences from the binary PCAP file.

Step 4: Deeper Analysis with tshark (Alternative Method)

While the flag was already found with `strings`, let's explore a more forensic approach using `tshark` to extract TCP payloads.

Notice that some TCP packets have a length of **42 bytes** (which is interesting). Let's extract all TCP packets with `len==42` and reconstruct their payloads:

```
tshark -r trace.pcap -T fields -e frame.time -e tcp.payload -Y "tcp.len==42" | sort |  
cut -f2 -d $'\t' | xxd -r -p
```

Output:

```
picoCTF{<REDACTED>}
```

The same flag appears!

Final Result

```
picoCTF{<REDACTED>}
```

Note: The actual flag is redacted here. In the real challenge, both `strings` and `tshark` will reveal the complete flag.

Key Concepts

Concept	Description
PCAP Analysis	Network packet captures contain raw packet data, including payloads that may contain plaintext flags.
Strings Extraction	A powerful technique for finding readable text embedded in binary files, even within network traffic.
Packet Payload	The actual data being transmitted, which may include flags, passwords, or other sensitive information.
TCP Payload Length	Filtering by <code>tcp.len</code> can help isolate packets containing specific data sizes.
Hexadecimal to Binary	Converting hex payloads back to raw bytes using <code>xxd -r -p</code> .

Pro Tips

- Always start with `strings`** – Before diving deep into packet analysis, a quick `strings` scan can reveal hidden flags immediately.
- Look for unusual packet lengths** – In this challenge, `tcp.len==42` was the key to finding the flag. Attackers often hide data in fixed-size packets.
- Combine `tshark`, `sort`, `cut`, and `xxd`** – This pipeline is powerful for extracting and reconstructing payloads from multiple packets.
- Don't overcomplicate** – Sometimes the simplest solution (like `strings`) is all you need!